

NETWORKS AND COMPLEXITY

Solution 12-1

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 12.1: Integration [Lvl. 1]

Find the general solutions for the following differential equations:

a.

$$\dot{x} = 4x$$

b.

$$\dot{x} = \frac{1}{2x}$$

c.

$$\dot{x} = \frac{x}{a+t}$$

d.

$$\frac{dc_k}{dk} = -\frac{2c_k}{k}$$

Solution

All of these can be integrated by separation of variables

a.

$$\frac{dx}{dt} = 4x \tag{1}$$

$$\frac{1}{x}dx = 4dt \tag{2}$$

$$\int \frac{1}{x}dx = \int 4dt \tag{3}$$

$$\log x = 4t + C \tag{4}$$

$$x = x_0 e^{4t} \tag{5}$$

where $x_0 = e^C$.

b.

$$\frac{dx}{dt} = \frac{1}{2x} \tag{6}$$

$$\int 2x dx = \int dt \tag{7}$$

$$x^2 = t + C \tag{8}$$

$$x = \pm \sqrt{t + C} \tag{9}$$

c.

$$\frac{dx}{dt} = \frac{x}{a+t} \quad (10)$$

$$\int \frac{1}{x} dx = \int \frac{1}{a+t} dt \quad (11)$$

$$\log(x) = \log(a+t) + C \quad (12)$$

$$x = e^{\log(a+t)+C} \quad (13)$$

$$x = A(a+t) \quad (14)$$

d.

$$\frac{dc_k}{dk} = \frac{-2c_k}{k} \quad (15)$$

$$\int \frac{1}{c_k} dc_k = \int \frac{-2}{k} dk \quad (16)$$

$$\log(c_k) = -2\log(k) + C \quad (17)$$

$$c_k = Ak^{-2} \quad (18)$$